

# Implementation and Feasibility Plan

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## 1. Feasibility today (what is running)

The prototype is built and tested: a FastAPI core, a website with voice and GPS, a Supabase database with a private audio bucket, and a Streamlit officer dashboard. It runs five services (water, electricity, roads, sanitation, general triage) on one engine. Everything used is on free tiers, and everything server-side is stateless except the database.

## 2. Phased plan

| Phase                            | Goal                       | Work                                                                                                                                                                                       | Depends on         | Exit check                                              |
|----------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------|
| 0. Prototype (done)              | Prove the flow             | Website chat, voice, routing, tickets, dashboard, status                                                                                                                                   | none               | Ten scenarios exercised; 415 + 35 tests pass            |
| 1. Make the data real            | Replace demo data          | Verified ward and office data for Bhopal (RTI or a visit to BMC); ward centre points from the ward boundary file; officer roles; resolve BMC-versus-PHED ownership                         | BMC cooperation    | GPS picks the correct ward; no (DEMO) rows in the pilot |
| 2. Harden for real users         | Safe to open to the public | Phone-verified complaints (SMS OTP); rate limits on /message and /speak ; per-department officer accounts; officer notification when a ticket arrives; retention policy and consent notice | Phase 1            | Abuse test passes; officers see only their department   |
| 3. Pilot in Bhopal               | Measure real impact        | Live pilot with a few wards, then all Bhopal wards; track reassignments, time to first status change and voice use                                                                         | Phase 2            | The four impact metrics in item 05 have real numbers    |
| 4. Add departments and districts | Scale across MP            | One YAML service spec plus office rows per department; a central department registry file with checks; another city as a second pilot                                                      | Phase 3            | A new department goes live with no engine code change   |
| 5. Integrate                     | Fit the existing ecosystem | Ticket hand-off to MPOnline's Customer Solutions Hub / CM Helpline; WhatsApp channel on the same engine; DigiLocker-based document help with citizen consent                               | MPOnline agreement | Tickets land in the existing system                     |

Phase order matters: phone verification and rate limiting come before any public launch, because an open SMS endpoint costs money per message and invites fake complaints.

## 3. Resources and cost (indicative, to be confirmed)

| Item                                  | Today                                                                                     | At pilot scale |
|---------------------------------------|-------------------------------------------------------------------------------------------|----------------|
| LLM (Groq free tier, Gemini fallback) | Free; Groq free tier is about 8,000 tokens per minute per model, so bursts can exhaust it | Paid tier      |
| Speech (Sarvam)                       | Free tier                                                                                 | Paid, per use  |

| Item                                                           | Today                                      | At pilot scale                                                                              |
|----------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------|
| Database and storage (Supabase)                                | Free; can pause after 7 days of inactivity | Paid plan                                                                                   |
| Hosting (Railway or Render, Vercel, Streamlit Community Cloud) | Free; free hosts sleep when idle           | Paid plan                                                                                   |
| SMS OTP for phone verification                                 | Not in the product; one test succeeded     | Per-message cost; Indian SMS needs DLT (sender and template) registration, which takes days |
| Exact figures                                                  | TBD                                        | TBD                                                                                         |

## 4. Risks and how we handle them

| Risk                            | Mitigation                                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| AI misroutes a complaint        | Validator against the spec; confidence threshold; <code>needs_review</code> ; officer reassign with a logged correction               |
| Provider outage or rate limit   | LLM chain of four models; ASR fallback; short timeouts; Hindi error messages                                                          |
| Spam or fake complaints         | Phase 2: phone verification, rate limits, duplicate detection                                                                         |
| Wrong or unverified office data | Verified-or-labelled rule: every unverified row is marked DEMO; a registry with source and verified fields is proposed                |
| Privacy and DPDP Act compliance | Minimal data by design; a full review (consent, retention, grievance officer) is a Phase 2 task, and we do not claim compliance today |
| Free tier limits                | Move to paid plans before any public pilot                                                                                            |

## 5. Team and process

A two-person team, working spec-first: every module has a written spec ( `docs/specs/S01` to `S31` ) and plan before code. The repository holds the specs, tickets and research so a new developer can pick it up ( `docs/ONBOARDING.md` ).